
Appendix A2

WebArAr User Guide

WebArAr: A new web-based platform for $^{40}\text{Ar}/^{39}\text{Ar}$ data reduction with insights on 3D K–Cl–Ar correction and uncertainty propagation

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Introduction

WebArAr is a web-based application based on Django designed to reduce $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology data. The backend functionalities are packaged into a module called ArArPy. Access through PyPi.

Visit WebArAr via: <https://www.webarar.net/>.

Documentation online: <https://www.webarar.net/doc/>.

1. Raw File Import

1.1 Explanation of Terms in Filter Editor

File Type Restrictions:

- **Extension:** Inputting file extension to match files.
- **File Type:** Dropdown menu including Text, Excel, Qtegra Exported XLS, and Seq.
- **Delimiter:** Dropdown menu including Tab, Semicolon, Space, Comma, and Other. When Other is selected, the content in the subsequent input box is used as the delimiter.

Isotope Reading Restrictions:

- **Multiple Experimental Steps:** Dropdown menu to determine if a single file contains multiple experimental steps (e.g., CSV files exported by Qtegra are multi-step files, while NGXDP files generated by Isotopx Isolinx are single-step files).
- **Data Sheet Index:** Numeric input box to specify the worksheet number containing isotope data (only valid for multi-sheet Excel files; set to 1 for others).
- **Header Rows:** Numeric input box for the number of header rows.
- **Data Direction:** Dropdown menu with *Vertical* (each cycle arranged downward) and *Horizontal* (each cycle arranged to the right) to specify the arrangement of isotope data.
- **Cycle Num:** Numeric input box for the number of cycles in a single stage of isotope data.

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- **Ar40 Intensity / Ar39 Intensity / Ar38 Intensity / Ar37 Intensity / Ar36 Intensity:** Numeric input boxes for the row and column positions of signal values for each argon isotope.
 - **Inlet Time (for each isotope):** Numeric input boxes for the row and column positions of measurement time for each argon isotope.
 - **Lines of Each Cycle:** Numeric input box for the number of rows occupied by data per cycle.
 - **Break Lines:** Numeric input box for the number of blank lines after each cycle.
 - **Intensity Unit:** Text input box for the display unit of isotope signals.
 - **Intensity Scale Factor:** Numeric input box to match display units with actual values (multiplicative factor).
 - **Lines of Each Step:** Numeric input box to specify the number of rows per stage in multi-step files.

Stage Information Restrictions:

- **Read Information from File Name:** Checkbox to determine if information is read from the file name (highest priority).
- **Filename Format:** Text input box for the file name format (e.g., "{en}-{sn}-{s}" matches "24WHN0100-01-12345", where "en" = experiment name (24WHN0100), "sn" = step name (01), and 12345 is a unique suffix with no information).
- **Parse information from a string:** Checkbox to determine if information is extracted from a specific line of string.
- **String Format:** Text input box for the format of the aforementioned string.
- **Zero Date in One String:** Checkbox to determine if the date is stored in a single string.
- **Date Format:** Text input box for the date string format.
- **Zero Time in One String:** Checkbox to determine if the time is stored in a single string.
- **Time Format:** Text input box for the time string format.
- **Time Zone:** Dropdown menu to select the time zone of the zero-time in the file.
- **Exp Name:** Experiment name.
- **Step Name:** Stage name.
- **Step Type:** Stage type (e.g., Unknown, Blank, Air).
- **Step Label:** Stage label (e.g., 10%, 100, 400°C).
- **Zero Date (Date/Year):** Zero-date or year index.
- **Zero Date (Mon.):** Zero-date month index.
- **Zero Date (Day):** Zero-date day index
- **Zero Time (Time/Hour):** Zero-time or hour index

Experimental Information Restrictions:

- **Zero Time (Min.):** Zero-time or minute index
- **Zero Time (Sec.):** Zero-time or second index
- **Sample Name / Sample Location / Sample Material:** Sample name, location, and material.
- **Experiment Type:** Experiment type (e.g., Crushing, Laser, Heating).
- **Sample Weight:** Sample weight.
- **Step Unit:** Stage unit.
- **Heating Time:** Heating time.
- **Instrument Name / Researcher / Analyst / Laboratory:** Instrument name, researcher, analyst, and laboratory.
- **J-value / J-value Error:** J-value and its error.
- **MDF value / MDF Error:** Mass discrimination factor (MDF) and its error.
- **Calc Params / Irradiation Name / Irradiation Label:** Calculation parameter name, irradiation parameter name, and irradiation label.
- **Irradiation H / X / Y:** Irradiation positions H, X, and Y.
- **Standard Name / Standard Age / Std Age Error:** Standard name, standard age, and its error.
- **J value and MDF errors in percentage:** Checkbox to determine if J-value and MDF errors are relative errors (%).
- **Standard age error in percentage:** Checkbox to determine if the standard age error is a relative error (%).

1.2 Filters Examples

(1) NGXDP Files Exported by NGX-600

NGXDP files are automatically generated by the ISOLINX software suite with for example the Isotopx NGX noble gas mass spectrometer during experiments, recording single-run results. They are characterized by: (1) Files are separated by headers such as #HEADER, #SAMPLELIST, #COLLECTORS, #BASELINES, #RESULT-RATIOS, #RATIO, #RESULT-ISOTOPES, #ISOTOPE, and #END; (2) #ISOTOPE records isotope signal intensities in the order of ^{36}Ar , ^{37}Ar , ^{38}Ar , ^{39}Ar , and ^{40}Ar , with each isotope's measurement cycle occupying one line, recorded sequentially. The example file contains 38 cycles.

Based on NGXDP file parsing, the filter can be defined as follows: Extension = ngxdp, File Type = Text, Delimiter = Comma, Non-multi-step experimental file, Data Direction = Vertical, Cycle Num = 38*, ^{40}Ar intensity starts at row 397, column 3 (397, 3), inlet time starts at row 397, column 2 (397, 2); ^{39}Ar intensity (356, 3), inlet time (356, 2); ^{38}Ar intensity (315, 3), inlet time (315, 2); ^{37}Ar intensity (274,

3), inlet time (274, 2); ³⁶Ar intensity (233, 3), inlet time (233, 2); each cycle occupies 1 row, isotope signal unit = mV, Intensity Scale Factor = 1000 (converting from fV to mV). The file name (25WHN0144-51-23565.NGXDP) consists of the experiment number, stage number, and auto-incrementing suffix. Thus, "Read Information from File Name" can be checked with the format "{en}-{sn}-{s}", where fields wrapped in {} are sample information types, and other content serves as separators to split the string. Zero-time is stored in row 13, column 2 as "06/08/2025 07:02:43.513" (containing both date and time). Therefore, both ZeroTime (Date) and ZeroTime (Time) are set to (1, 13, 2), with "Zero Date in one String" and "Zero Time in one String" checked, and the format set to "%d/%m/%Y %H:%M:%S.%f". Using the experiment list editing function in Isotopx NGX software, as much content as possible can be carried in raw data to enhance automatic reading richness. For example, the operator can input the experimenter's name, irradiation number, original sample number, sample type, J-value, etc., in the Operator field (Fig. 1) and fill the corresponding indices into the filter.

**Note: The number of cycles can exceed 38 in this example. When reading the 39th row, an error will occur when identifying isotopes (as the 39th row is blank), and the loop for reading the current isotope will exit.*

Run	Sample No	Filename	Power Temp	Baseline Filename	Analysis Type	Operator
1	1	25WHN0065-1	Blk	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
2	1	25WHN0065-2	5	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
3	1	25WHN0065-3	6	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
4	1	25WHN0065-4	7	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
5	1	25WHN0065-5	8	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
6	1	25WHN0065-6	9	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
7	1	25WHN0065-7	10	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
8	1	25WHN0065-8	Blk	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
9	1	25WHN0065-9	11.5	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
10	1	25WHN0065-10	13	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
11	1	25WHN0065-11	14.5	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
12	1	25WHN0065-12	16	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
13	1	25WHN0065-13	18	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
14	1	25WHN0065-14	20	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
15	1	25WHN0065-15	Blk	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
16	1	25WHN0065-16	22	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
17	1	25WHN0065-17	24	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
18	1	25WHN0065-18	26	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
19	1	25WHN0065-19	29	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
20	1	25WHN0065-20	32	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
21	1	25WHN0065-21	35	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
22	1	25WHN0065-22	Blk	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
23	1	25WHN0065-23	38	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
24	1	25WHN0065-24	42	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
25	1	25WHN0065-25	46	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
26	1	25WHN0065-26	50	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
27	1	25WHN0065-27	55	EBL2503-18508	Analysis	BAI XJ, WH05-N18, D2030-1, 石英, 0.112928265, 25 李益龙
28	1	25WHN0065-28	60	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
29	1	25WHN0066-01	Blk	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
30	1	25WHN0066-2	5	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
31	1	25WHN0066-3	6	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
32	1	25WHN0066-4	7	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
33	1	25WHN0066-5	8	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
34	1	25WHN0066-6	9	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
35	1	25WHN0066-7	10	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
36	1	25WHN0066-8	Blk	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
37	1	25WHN0066-9	11.5	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
38	1	25WHN0066-10	13	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
39	1	25WHN0066-11	14.5	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
40	1	25WHN0066-12	16	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙
41	1	25WHN0066-13	18	EBL2503-18508	Analysis	BAI XJ, WH05-N10, D0560-1, 石英, 0.115952545, 25 李益龙

Fig. 1 Isolinx user-interface showing sample information embraced in the column *operator*

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Fig. 3 An example of the NGXDP files

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CSV files are one of the optional formats for single-step files exported by Qtegra. They are semicolon-separated text files that can contain various parameters based on different configuration choices. In the example file, the first four rows are headers, and data from the fifth row onward includes test information and results for each stage (one stage per row). The file records data from nine detectors: the first nine columns are the extrapolated zero-time means calculated by the software, followed by signals for each cycle (titled MainRuns), then Qtegra's built-in extrapolation results (e.g., Extrapolation.ResultCycleValues, Extrapolation.ResultStandardError) which are not of interest, and finally experimental information including start/end test times, instrument settings, and user-defined

content. Please note that the content of the csv file output varies under different export settings.

Accordingly, the filter is set as follows: Extension = csv, File Type = Text, Delimiter = Semicolon,

Multi-step experimental file, Header Rows = 4, Data Direction = Horizontal, Number of stages = 15.

Select the correct isotope data from the nine detectors based on actual conditions: in this example, ^{40}Ar

is detected by the Faraday cup at position H2 (intensity at (1, 33), measurement time at (1, 34)), ^{39}Ar by

the CDD detector at H1 (intensity at (1, 53), measurement time at (1, 54)), ^{38}Ar by the CDD detector at

AX (intensity at (1, 48), measurement time at (1, 49)), ^{37}Ar by the CDD detector at L1 (intensity at (1,

43), measurement time at (1, 44)), and ^{36}Ar by the CDD detector at L2 (intensity at (1, 38), measurement

time at (1, 39)). Each cycle occupies 45 columns with no blank rows/columns, signal unit = fA, Intensity

Scale Factor = 1, and each stage occupies 1 row. The experiment name is read from (1, 5, 2), stage name

from (1, 5, 1), stage label from (1, 5, 874), and zero-time from (1, 5, 859). For multi-step files, each

stage's data occupies a different row by default, and these parameters are read sequentially based on the

number of rows per stage (e.g., zero-time for the first stage is (1, 5, 859), so zero-time for the second

stage is $(1, 5+1 \times 1, 859)$, where 1×1 indicates 1 previous stage occupying 1 row).

[illegible][illegible]

Fig. 4 An example of CSV files exported by Otegra with Helix MC

● Edit an RawDataInputFilter Project ★Qtetra-CSV ▼
○ Create New

Format

Extension: File Type: ▼

Delimiter:

Isotopes

☒ Multiple Experimental Steps

Data Sheet Index: Header Rows:

Data Direction: Cycle Num:

⁴⁰Ar Intensity: Inlet Time:

³⁹Ar Intensity: Inlet Time:

³⁸Ar Intensity: Inlet Time:

³⁷Ar Intensity: Inlet Time:

³⁶Ar Intensity: Inlet Time:

Lines of Each Cycle: Break Lines:

Intensity Unit: Intensity Scale Factor:

Lines of Each Step:

Parse information for each step

Three inputs are Sheet Index, Row Index, and Column Index. Only integer or zero will be allowed.

☐ Read Information from File Name

Filename Format:

☐ Parse information from a string

String Format: ⓘ

☒ Zero Date in one String

Date Format: ⓘ

☒ Zero Time in one String

Time Format: ⓘ

Time Zone: ⓘ

Exp Name: Step Name:

Step Type: Step Label:

ZeroTime (Date): ZeroTime (Time):

ZeroTime (Mon.): ZeroTime (Min.):

ZeroTime (Day): ZeroTime (Sec.):

Experimental information (optional)

Optional information of the given sample, which will be recorded into the final sample instance.

Sample Name: Sample Location:

Sample Material: Experiment Type:

Sample Weight: Step Unit:

Heating Time:

Instrument Name: Researcher:

Analyst: Laboratory:

J-value: J-value Error:

MDE value: MDE Error:

Calc Params: Irradiation Name:

Irradiation Label: Irradiation H:

Irradiation X: Irradiation Y:

Standard Name:

Standard Age: Std Age Error:

☐ J value and MDF errors in percentage

☐ Standard age error in percentage

Fig. 5 An example of filter settings for Qtetra exported CSV files

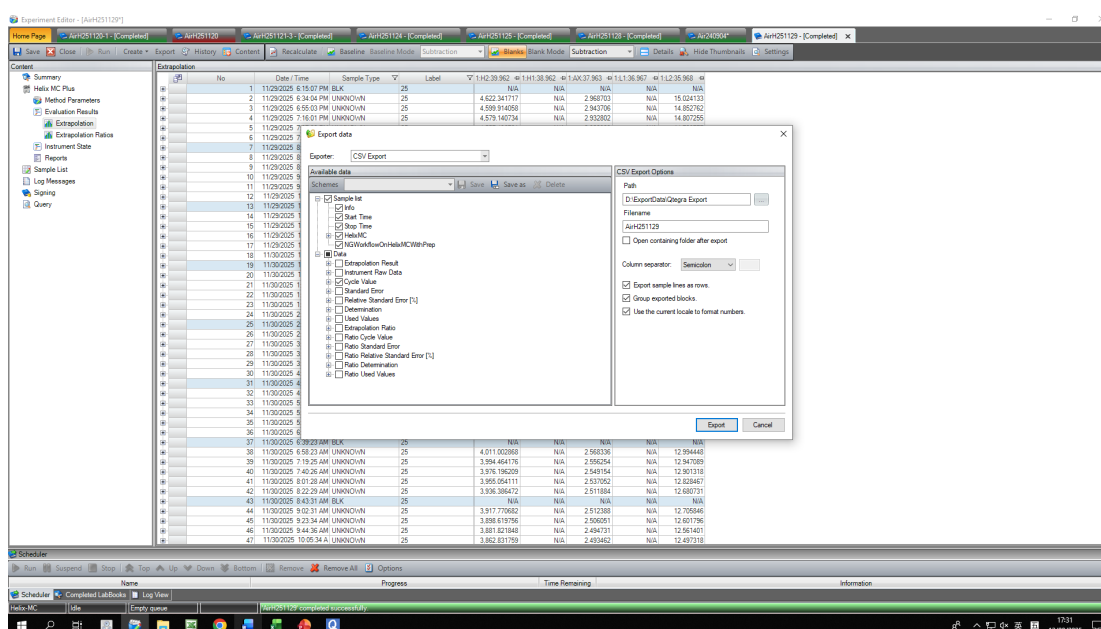


Fig. 6 User-interface of Qtetra. The content of the csv file varies under different export settings.

(3) Excel (XLS) Files Exported via Qtetra

In the Qtetra main interface, exporting to Excel via the right-click menu can export all test stages into a single XLS file (Fig. 7). Each stage's data occupies 55 rows, including headers, isotope signal values, extrapolated zero-time values, and thumbnails (Fig. 8).

For this file, the filter could be set as follows: Extension = xls, File Type = Excel, No delimiter required for Excel files, Multi-step experimental file, Header Rows = 0, Data Direction = Vertical, Number of stages = 10. The example file measures five isotopes simultaneously in a single run: measurement time for all isotopes is (8, 5), and ⁴⁰Ar to ³⁶Ar intensities are located at (8, 6) to (8, 10). Each cycle occupies 1 row with 3 blank rows (or set to 4 rows per cycle with 0 blank rows), signal unit = fA, Intensity Scale Factor = 1, and each stage occupies 55 rows. The file name (for example,

157 19WHA0102.xls) could be experiment name. Zero-time is read sequentially from (1, 2, 2). The example
 158 file uses a 12-hour timestamp format "%m/%d/%Y %I:%M:%S %p". Stage name is (1, 2, 1), stage type
 159 (1, 2, 3), and stage label (1, 2, 4).

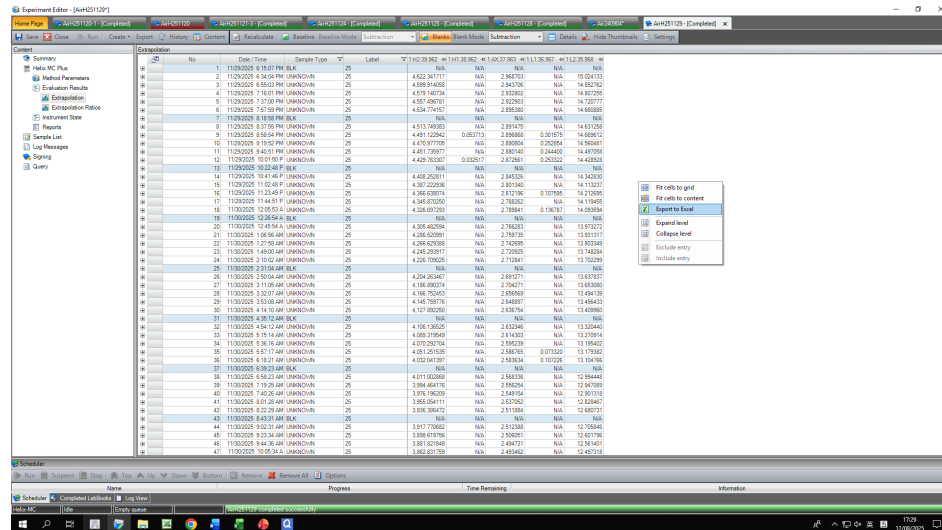


Fig. 7 Right-click to export excel files with Qtegra

	A	B	C	D	E	F	G	H	I	J	K
1	No	Date / Time	Sample Type	Label	1:H2:39.962	1:H1:38.964	1:AT:37.95	1:11:36.967	1:12:33.968		
2		6/11/2019 7:03:59 PM BLK			N/A	N/A	N/A	N/A	N/A		
3											
4											
5											
6											
7											
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56	No	Date / Time	Sample Type	Label	1:H2:39.962	1:H1:38.964	1:AT:37.95	1:11:36.967	1:12:33.968		
57		6/11/2019 7:15:29 PM UNKNOW			94273.86558	48.9158543	135.4465098	0.133481344	181.7194063		
110											
111	No	Date / Time	Sample Type	Label	1:H2:39.962	1:H1:38.964	1:AT:37.95	1:11:36.967	1:12:33.968		
112		6/11/2019 7:28:13 PM UNKNOW			73700.88008	44.40007989	107.8793042	0.209637762	117.8333800		
165											
166	No	Date / Time	Sample Type	Label	1:H2:39.962	1:H1:38.964	1:AT:37.95	1:11:36.967	1:12:33.968		

Fig. 8 Data structure of excel files exported from Qtegra

(4) AHD files

XLS files exported by Qtegra need to be split into single-step files for correct reading by ArArCALC. AHD is a splitting method adopted by the $^{40}\text{Ar}/^{39}\text{Ar}$ laboratory at Vrije Universiteit Amsterdam, Netherlands. AHD files are Tab-separated text files (Fig. 9) containing information such as sample name (actually the experiment name with stage number), experiment name, project name, irradiation details, standard, instrument, timestamp, personnel, temperature, and J-value.

1	Sample	11_24FY49a
2	Experiment	24FY49a
3	Project	
4	Irradiation	
5	Standard	
6	Instrument	Helix IE-13
7	Time Stamp	30/06/2024 23:43:16
8	Analyst	
9	Temperature	70 Laser
10	J-Value	
11	Fractionation	
12	Volume Correction	1
13	Counters	1 15 1 0
14		
15	Time	Intensity 37 38 39 40 Cycle # Peakreading #
16	393.930784	0.030478093
17	393.930784	0.040538852
18	393.930784	0.077091196
19	432.698784	5.855806454
20	393.930784	16.94827518
21	471.522784	0.030525842
22	471.522784	0.040242807
23	471.522784	0.076479996
24	510.298784	5.80031938
25	471.522784	16.91259362
26	549.138784	0.030258447
27	549.138784	0.039746214
28	549.138784	0.076336746
29	507.911784	5.838256726
30	549.138784	16.88581563
31	626.707784	0.030005376
32	626.707784	0.04036218
33	626.707784	0.076589821
34	665.451784	5.733479616
35	626.707784	16.83527995
36	704.259784	0.030669089
37	704.259784	0.04034308
38	704.259784	0.074488824
39	743.028784	5.699339367
40	704.259784	16.80055135
41	781.805784	0.030597465
42	781.805784	0.039856037
43	781.805784	0.073667526
44	820.564784	5.627561863
45	781.805784	16.82713655
46	859.360784	0.030568816
47	859.360784	0.040457678
48	859.360784	0.073366701
49	898.140784	5.580580514
50	859.360784	16.76919293
51	936.893784	0.030525842

Fig. 9 An example of AHD files

For this example, the filter could be defined as follows: Extension = ahd, File Type = Text, Delimiter = Tab, Non-multi-step experimental file, Data Direction = Vertical, Cycle Num = 15, Header Rows = 15, ⁴⁰Ar intensity starts at row 20, column 2 (recorded as (5, 2) with 15 header rows), inlet time starts at row 20, column 1 (5, 1); similarly, ³⁹Ar intensity (4, 2), inlet time (4, 1); ³⁸Ar intensity (3, 2), inlet time (3, 1); ³⁷Ar intensity (2, 2), inlet time (2, 1); ³⁶Ar intensity (1, 2), inlet time (1, 1); each measurement cycle occupies 6 rows, isotope signal unit = mV, Intensity Scale Factor = 1. The file name (for example, 11_24FY49a_combined.ahd) consists of the stage number, experiment number, and suffix, with the stage name and experiment number identified using the format "{sn}_{en}_combined". Zero-time date and time are recorded in row 7, column 2 and row 7, column 3, so "Zero Date in one String" and "Zero Time in one String" are checked with formats "%d/%m/%Y" and "%H:%M:%S", respectively. Temperature can be used as the stage label (1, 7, 3), experiment type = Laser (recorded at (1, 9, 3)), and instrument name = Helix (recorded at (1, 6, 2)).

When argon isotopes are measured without peak jumping, each cycle of the five isotopes has the same measurement time, forming files with another structure (Fig. 10). The difference is that each cycle occupies one row, and isotope data are in different columns. Data Direction remains Vertical: measurement time is at (1, 1), and ³⁶Ar to ⁴⁰Ar intensities are at (1, 2) to (1, 6). Stage type (Blank or Unknown) is read from the experiment name 24RL5a at position (2, 2) (default = Unknown). For 24BRL5a, "B" in "ST" is correctly identified as a Blank step.

Raw File Filter

Step 1: Upload your files
Choose files No file chosen
Step 2: Select a filter for each uploaded file

☒	No.	Original Name	Sanitized Filename	Filter	Operation
☒	1	25WHN0144-3-23517	17653559778454491efact812327d840.NGXDP	*NGX-NGXDP	Remove
☒	2	11_24FY49a_combined	1765355985197914010361b6d1c0b557.ahd	*Otegra-AHD	Remove
☒	3	20241212_24FY80a	1765355990712537dadd31b5aa5cf67.csv	*Otegra-CSV	Remove
☒	4	19WHA0102	17653559986021343c5624c55c4398f1.xls	*Otegra-Export-to-Excel	Remove
☒	5	19WHA0102-1.0 et al	17653560029182635750645ccb37efae.seq	*WebAr-Ar-Seq	Remove

Close Submit

Fig. 13 Select filters and submit raw data files

1.4 Zero-Time Extrapolation

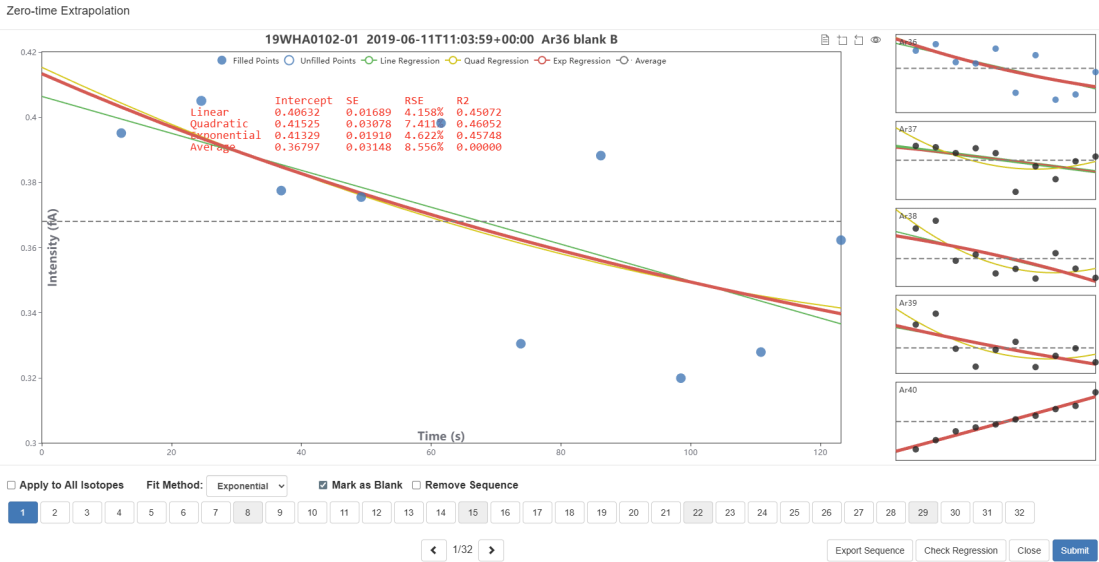


Fig. 14 Zero-Time extrapolation interface

The zero-time extrapolation page mainly displays fitting plots of inlet time and measured values for the five argon isotopes across different steps (Sequences). Click the five small plots on the right to switch the display of the large central plot (or use the up/down keys on the keyboard). The top of the plot shows a legend; click to toggle the display of each element. Solid scatter points are used for fitting calculations; click a scatter point to include or exclude it from fitting. Red text shows a comparison of intercept values, errors, and fitting correlation coefficients for different fitting methods, with the bold fitting line indicating the currently selected method. When clicking a scatter point, if "Apply to All isotopes" is checked, the selection/exclusion will apply to all five isotopes of the current step. "Fit method" is used to select different fitting methods to obtain intercept values for subsequent calculations (default: linear fitting for ³⁷Ar, exponential fitting for other isotopes). "Mark as Blank" sets the current step as a Blank step, and "Remove Sequence" removes the current stage from subsequent calculations. "Export Sequence" exports

Blank Correction

Fixed Values: Pre-run subtraction

☒	No.	Label	Unknown	Blank
☒	1	2.1	25WHN0144-04	25WHN0144-03
☒	2	2.2	25WHN0144-06	25WHN0144-05
☒	3	2.3	25WHN0144-07	25WHN0144-05
☒	4	2.4	25WHN0144-09	25WHN0144-08
☒	5	2.5	25WHN0144-10	25WHN0144-08
☒	6	2.6	25WHN0144-11	25WHN0144-08
☒	7	2.7	25WHN0144-13	25WHN0144-12
☒	8	2.8	25WHN0144-14	25WHN0144-12
☒	9	2.9	25WHN0144-15	25WHN0144-12
☒	10	3	25WHN0144-17	25WHN0144-16
☒	11	3.1	25WHN0144-18	25WHN0144-16
☒	12	3.2	25WHN0144-19	25WHN0144-16
☒	13	3.3	25WHN0144-21	25WHN0144-20
☒	14	3.4	25WHN0144-22	25WHN0144-20
☒	15	3.5	25WHN0144-23	25WHN0144-20
☒	16	3.6	25WHN0144-25	25WHN0144-24
☒	17	3.7	25WHN0144-26	25WHN0144-24
☒	18	3.8	25WHN0144-27	25WHN0144-24
☒	19	3.9	25WHN0144-29	25WHN0144-28
☒	20	4	25WHN0144-30	25WHN0144-28

Input Sequences:

Blank Sequence, splits sequences with semicolons (;)

New Blank:

Average()

Interpolate()

Add to List

Add Empty Blank

Import from Files

Blank Sequences:

25WHN0144-03

25WHN0144-05

25WHN0144-08

25WHN0144-12

25WHN0144-16

25WHN0144-20

25WHN0144-24

25WHN0144-28

25WHN0144-32

25WHN0144-36

25WHN0144-40

Last

1/5

Next

Close

The blank selection page is used to assign Blank steps to unknown sample steps. The left table displays each sample step (Unknown) and its corresponding Blank step. The checkbox at the top of the table toggles the mode for automatically searching Blank steps (default: "Pre-run", which finds the previous Blank before Unknown; other options: "Post-run", Blank after Unknown, "Adjacent", nearest Blank, and "Interpolation", interpolated Blank, which is only valid when an interpolation done). The left side of the page is for blank calculations: "Input" and "Output" are used to enter the Blank step name and the new stage name after calculation, respectively. "Average()" calculates the average of the input Blanks, "Interpolate()" computes the interpolation based on the input Blanks, "Add to List" adds the calculated Blank to the Blank list, "Add Empty Blank" adds an empty Blank to the list, and "Import from Files" reads Blank stages from other files.

Click a Blank name to add it to the input box, then click "Interpolate()" to enter the blank interpolation interface. Circular scatter points represent Blanks used for calculation, and square scatter points indicate the calculated values of both Blanks and unknowns. Click "Apply" to generate a new Blank. If the Output input box is left empty, the new stage is automatically named "Interpolated Blank"; otherwise, it uses the input name. Click "Add to List" to add the new stage to the Blank list. Select "Interpolated Blank" from the dropdown menu to use interpolated blank, or select "Interpolation Subtraction" from the top dropdown menu to apply interpolated blank subtraction to all steps.

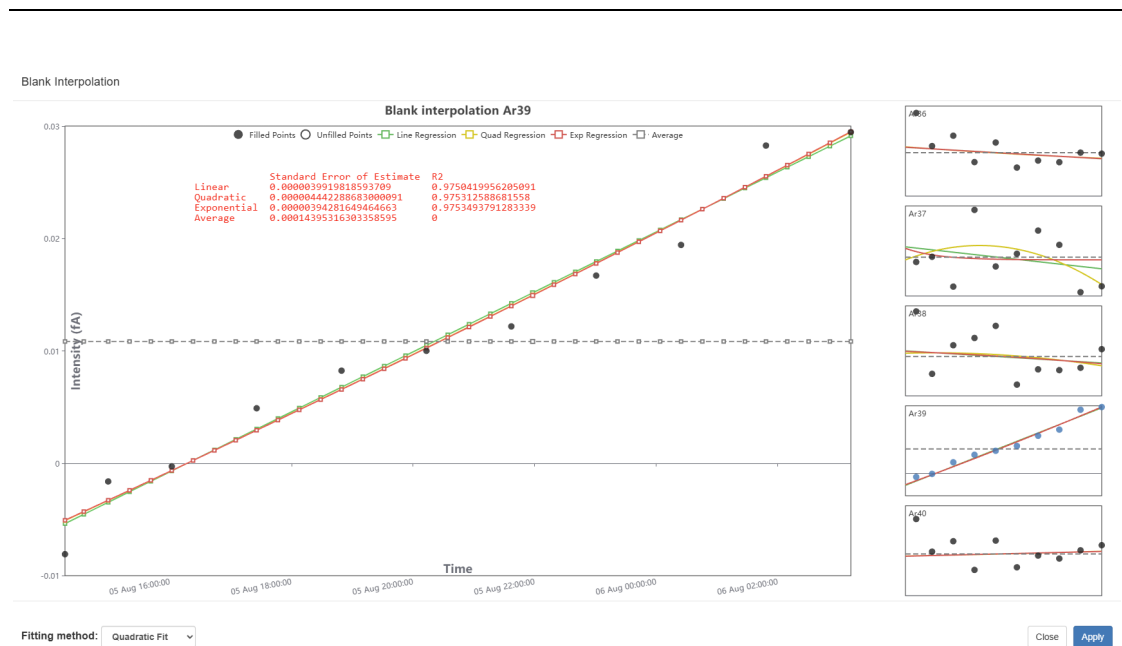


Fig. 15 Blank interpolation interface

1.6 Irradiation Parameter Selection

Select the required parameter set from the dropdown menu, and it is possible to edit any parameters. Irradiation parameter sets are defined in the "Irradiation Params" module on the main interface, mainly including interference nuclear reaction correction factors and irradiation duration.

Irradiation Parameters

☒ Use an Irradiation Project ☐ Use Input Params

☒ G220MY

All uncertainties are required to be stated at the 1 σ level.

Correction Params	
$^{40}\text{Ar}/^{39}\text{Ar}$ Trapped	298.56 $\pm$ 0 %s
$^{40}\text{Ar}/^{39}\text{Ar}$ Cosmogenic	0.018 $\pm$ 35 %s
$^{38}\text{Ar}/^{39}\text{Ar}$ Trapped	0.1885 $\pm$ 0 %s
$^{38}\text{Ar}/^{39}\text{Ar}$ Cosmogenic	1.493 $\pm$ 3 %s
$^{36}\text{Ar}/^{37}\text{Ar}$ Ca	0.00061748 $\pm$ 0 %s
$^{36}\text{Ar}/^{37}\text{Ar}$ Ca	0 $\pm$ 0 %s
$^{36}\text{Ar}/^{37}\text{Ar}$ Ca	0.00023477 $\pm$ 0 %s
$^{40}\text{Ar}/^{39}\text{Ar}$ K	0.0023228 $\pm$ 0 %s
$^{38}\text{Ar}/^{39}\text{Ar}$ K	0.0094194 $\pm$ 0 %s
$^{36}\text{Ar}/^{39}\text{Ar}$ Cl	0 $\pm$ 0 %s
$^{36}\text{Cl}/^{39}\text{Cl}$ Productivity	262.8 $\pm$ 0.65 %s
K/Ca	0.387 $\pm$ 0 %s
K/Cl	0 $\pm$ 0 %s
Ca/Cl	0 $\pm$ 0 %s

Irradiation Cycles	
Cycle Number	2
End Time	15/10/2014 12:47
Duration	18.22 hour(s)
End Time	16/10/2014 16:55
Duration	12 hour(s)

Irradiation Info	
Name	G220MY
Location	Manyang
Information	Manyang

Last 2/5 Next Close

Fig. 16 Irradiation parameter selection

1.7 Calculation Parameter Selection

Select the required parameter set from the dropdown menu. Calculation parameter sets are defined in the "Calculation Params" module on the main interface, mainly including constants such as decay constants and atomic masses.

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Sample & Experiment Info

Information irrelevant to the calculation.

Experiment Info

Experiment Name

25WHN0144

Experiment Type

Blank-LaserB-500s

Instrument

Laboratory

Laboratory More

more information

Analyst

BAI XU

Sample Info

Sample Name

DQ315-3

Type

Unknown

Sample Material

伊利石

Sample Weight

Sample Location

Researcher

25 美国

Irradiation Info

Irradiation Label

WH05B-H2

Irradiation H

Irradiation X

Irradiation Y

Last

5/5

Submit

Close

Fig. 19 Additional sample and experimental information

2. Tables and Figures

Information

Unknown

Blank

Corrected

Degas Pattern

Publish

Age Spectra

Isochrons

Total Params

Age Spectra

Non Isochron

Inv Isochron

K-Ci-Ar 1

K-Ci-Ar 2

K-Ci-Ar 3

3D Correlation

Degas Pattern

Ages Distribution

20241203_24FY10a

Experiment Name

20241203_24FY10a

Experiment Type

Sample Name

20241203_24FY10a

Sample Type

Unknown

Material

Muscovite

Location

Higher Himalayan Lusha Leucogranite

Researcher

Yang

Laboratory

VUJ

More

VUJ25-F10 132117020

Analyst

Yang

Sample Photo

2024/12/17 22:54:16 Information

Save Changes

Info Params

Calc Params

Strip Params

Recalculate

Export

Open: 2024/12/17 18:32:18

Fig. 20 Object view of WebArAr showing tables and figures

2.1 Tables

- **Information:** Sample information, such as name, mineral material, laboratory, etc.
- **Unknown:** Isotopic values for each sample sequence.
- **Blank:** Used blank values of each sample sequence
- **Corrected:** The isotopic values of each sample sequence after necessary corrections, including blank correction, mass discrimination correction, and decay correction
- **Degas Pattern:** The value of each isotope after distinguishing their sources

- **Publish:** In most cases, it is used to publish Ar-Ar data in a paper, including step name, step label, corrected argon isotopic values, apparent age, age error, proportion of $^{40}\text{Ar}_r$ in each sequence, ratio of $^{39}\text{Ar}_K$ released in each sequence to the total amount, and Ca/K value.
- **Age Spectra:** $^{40}\text{Ar}_r/^{39}\text{Ar}_K$ and apparent ages.
- **Isochrons:** Data of normal, inverse, and CI-related plots and the 3D plot.
- **Total Params:** All parameters involved in calculation.

2.2 Figures

- **Age Spectra:** Age spectra.
- **Nor. Isochron:** Normal isochron plot.
- **Inv. Isochron:** Inverse isochron plot.
- **K-CI-Ar 1:** CI correlation plot 1 in two dimensions. Note the difference in axes between three CI plots.
- **K-CI-Ar 2:** CI correlation plot 2 in two dimensions.
- **K-CI-Ar 3:** CI correlation plot 3 in two dimensions.
- **3D Correlation:** Correction plot in three dimensions.
- **Degas Pattern:** Shows the proportion of different isotope contributions released at each sequence.
- **Ages Distribution:** Shows the distribution of apparent ages, including bar charts and KDE curves

3. Set Parameter Sets

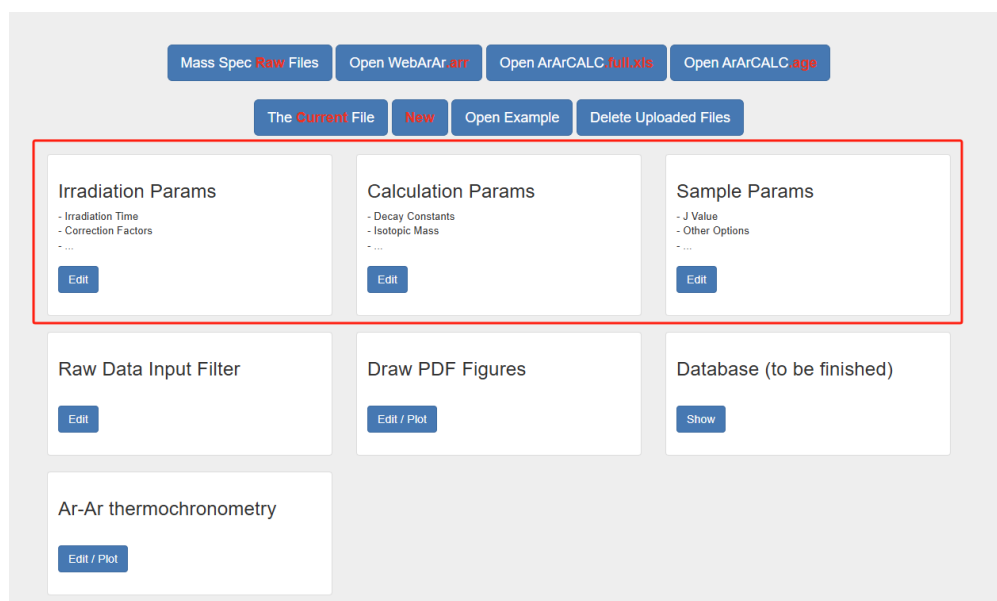


Fig. 21 Entrances of editing parameter sets

The screenshot shows the 'Calculation Parameters' web application. A modal window is open for creating a new parameter set. The modal has three input fields: 'Params Name', 'Code', and 'Email address'. Below these fields are 'Cancel' and 'OK' buttons. The background interface is partially visible, showing a table of decay constants and activities.

Decay Constants			
^{40}K	5.5306×10^{-10}	± 0.27	%/yr [1/yr]
$^{40}\text{K}(\text{EC})$	5.7579×10^{-11}	± 0.278	%/yr [1/yr]
$^{40}\text{K}(\beta_-)$	4.9449×10^{-10}	± 0.27	%/yr [1/yr]
$^{40}\text{K}(\beta_+)$	0	± 0	%/yr [1/yr]
^{39}Ar	2.946×10^{-7}	± 0.558	%/yr [1/yr]
^{37}Ar	0.000823	± 0.143	%/yr [1/yr]
^{36}Cl	0.00002257	± 0.65	%/yr [1/yr]

Decay Activities			
^{40}K	31.58	± 0.2027	%/yr
$^{40}\text{K}(\text{EC})$	3.31	± 1.209	%/yr
$^{40}\text{K}(\beta_-)$	28.27	± 0.177	%/yr
$^{40}\text{K}(\beta_+)$	0	± 0	%/yr

Other constants			
^{23}Ar Mass	35.96774628	± 0	
^{27}Ar Mass	36.96677709	± 0	
^{37}Ar Mass	37.96277302	± 0	
^{39}Ar Mass	38.964313	± 0	
^{40}Ar Mass	39.962383123	± 0	
K Mass	39.0983	± 0	
NA	6.0221367×10^{23}	± 0.000059	%/yr
Year	31556926	± 0	%/yr
$^{40}\text{K}/\text{K}$	0.000117	± 0.855	%/yr
$^{23}\text{Cl}/^{37}\text{Cl}$	3.0856	± 2	%/yr
HCl/Cl	0.2	± 20	%/yr
$^{40}\text{Ar}/^{39}\text{Ar}(\text{at})$	298.36	± 0.1	%/yr
$^{36}\text{Ar}/^{39}\text{Ar}(\text{at})$	0.1880	± 0.001092	%/yr

Fig. 22 Creating a calculation parameter set

To create a calculation parameter set, param name, code and email address are required. The Code is only for identifying users when editing and deleting in the later.

4. Calculation/Recalculation

Recalculation will be not always applied automatically after setting new parameters. So the manually launch of a recalculation is recommended. The options that can be used:

- **Reset Arr Attributes:** Check the structure of the current object, which can be used to fix old files that are missing some attributes due to updates.
- **Recalculate Correction and Degass:** For correct recalculation, specific procedures need to be selected. For example, if you modify some blank values, you need to select all the options from blank correction to age calculation, otherwise the new parameters will not really affect the final result; If you just modify the J-value, you only need to select recalculation of apparent age, the previous stage is not affected by the J-value, and you should also **Reset Plot Data** to update plateau and isochron data.
- **Reset Plot Style:** reset styles of figures

The recalculation interface is divided into three columns: the first column displays executable recalculation operations, the second column shows the parameter conditions required for each operation, and the third column presents the progress of the recalculation. A progress bar at the top indicates the overall running progress. When an error occurs, the relevant parameters will be highlighted in red, and an error message will be displayed to prompt potential issues. Some errors are because of the invalid values of the current object. Some might be caused by the potential bugs of WebArAr, if so please contact the author.

Recalculation Options

Unselect All

60%

☒ 1. Reset Arr Attributes	arr version check	Completed: re_initial
☒ 2. Correction Blank	unknowns blanks gain-factors	Completed: re_corr_blank
☒ 3. Correction Mass Discr	blank corrected masses MDF method	Completed: re_corr_massdiscr
☒ 4. Correction Decay	Ar37 λAr37 Ar39 λAr39 irradiation cycles experiment datetime	Completed: re_corr_decay
☒ 5. Degas Ca	Ar37Ca 36/37Ca 38/37Ca 39/37Ca	Completed: re_degas_ca
☒ 6. Degas K	Ar39 Ar39Ca 38/39K 40/39K	Completed: re_degas_k
☒ 7. Degas Cl	Ar36 Ar38 38/36t 36/38Cl productivity λCl36 standing time	Completed: re_degas_cl
☒ 8. Degas atm	Ar36 38/36t 40/36t	Completed: re_degas_atm
☒ 9. Degas r	Ar40 Ar40K Ar40a	Completed: re_degas_r
☒ 10. Calculte Ratio	corrected values degas values K/Ca factor normalization factor for Monte Carlo: unknowns blanks mass gain mdf decay consts interference params experiment datetime	Monte Carlo simulating: 3/108
☒ 11. Calculate Apparent Ages	sample type	...
☒ 12. Reset Plot Data	spectra normal inverse K-CI-Ar 1 K-CI-Ar 2 K-CI-Ar 3 3D degas pattern age distribution	...
☒ 13. Reset Plot Style	plot data	...
☒ 14. Reset Table Data	table data step num	...
☒ 15. Reset Table Style	header col types	...

Apply

Close

Fig. 23 Recalculation options

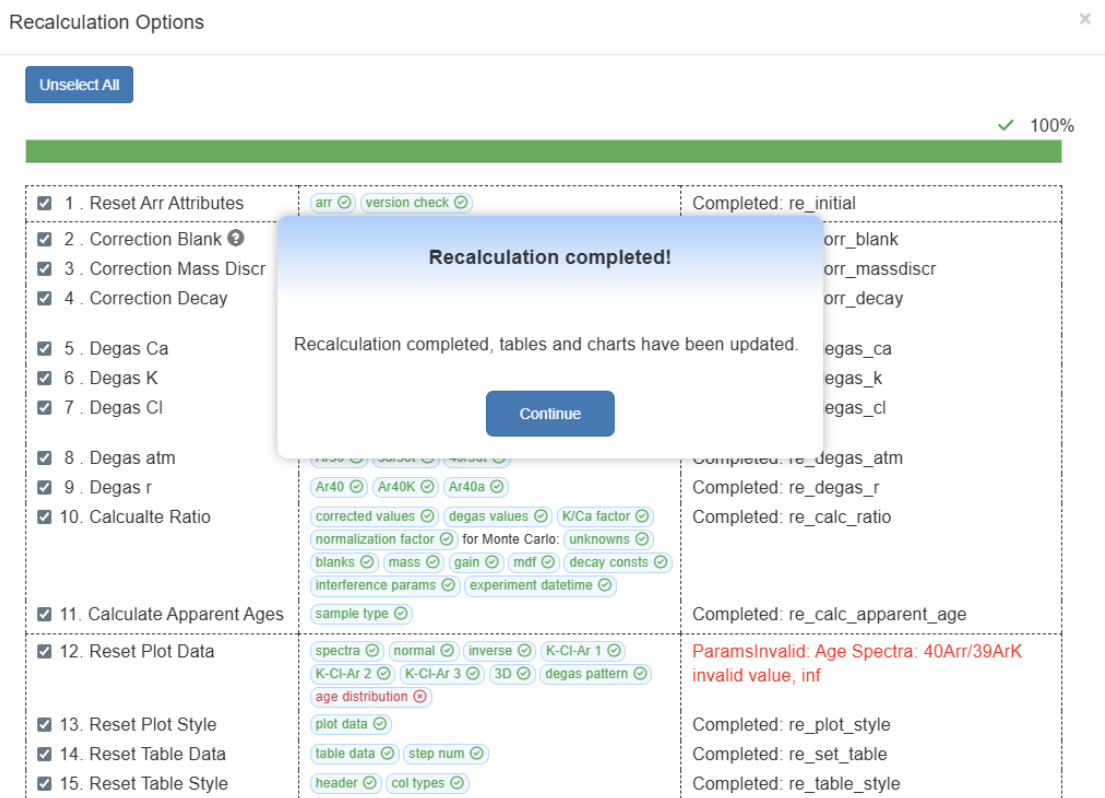
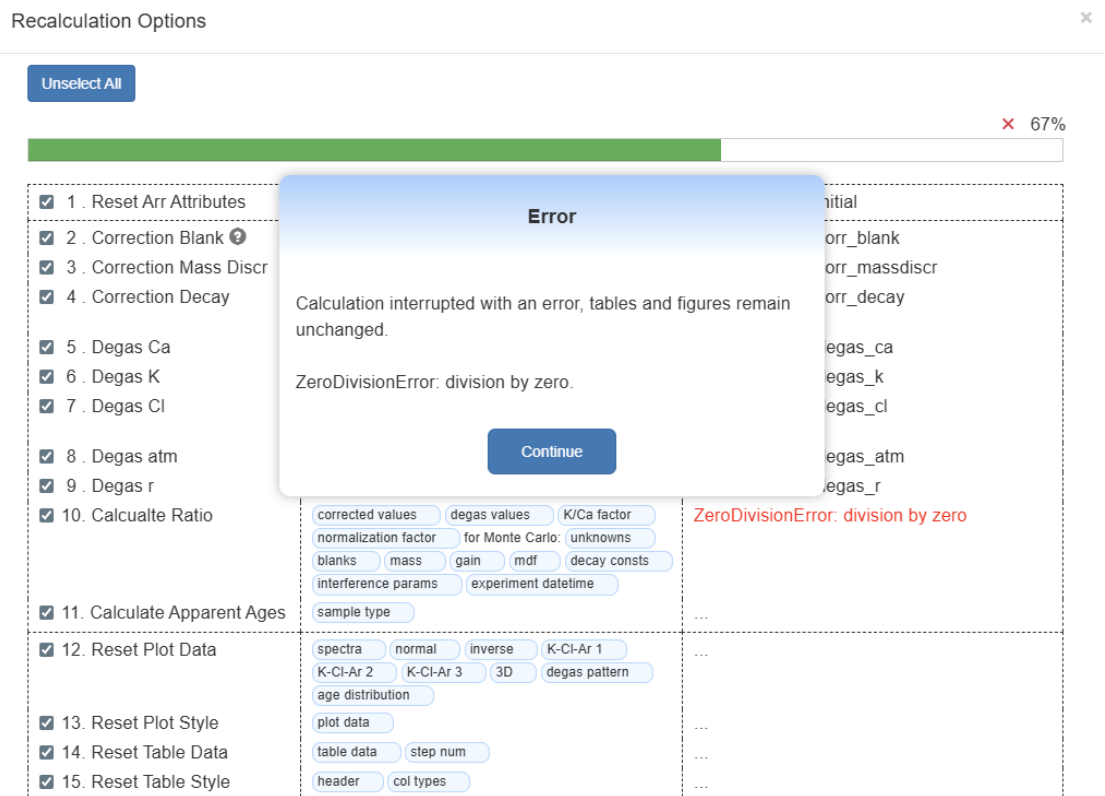


Fig. 24 Errors in recalculating

5. Isochron scatter points selection

The isochron supports two sets of point selection (Set1 and Set2), and the corresponding age plateau will be drawn in the age spectrum (the age plateau will be deducted for the air argon by the specified initial value according to the setting). The normal and inverse isochrons and plateau ages of the two point groups are displayed on the right.

Clicking a scatter allows user to select or deselect a point for regression. Regression and age are automatically recalculated with each click, and response speed is related to network speed and number of steps.

6. Change figure styles

Click the Style button at the bottom to set styles of the currently displayed figure, such as axes range, line width, color, size, etc. When the Style dialog is open, click on elements, such as lines, points, or texts, to set styles of the corresponding elements. For example, after opening Style, click a Scatter to open scatter style Settings, as shown in the following figure.

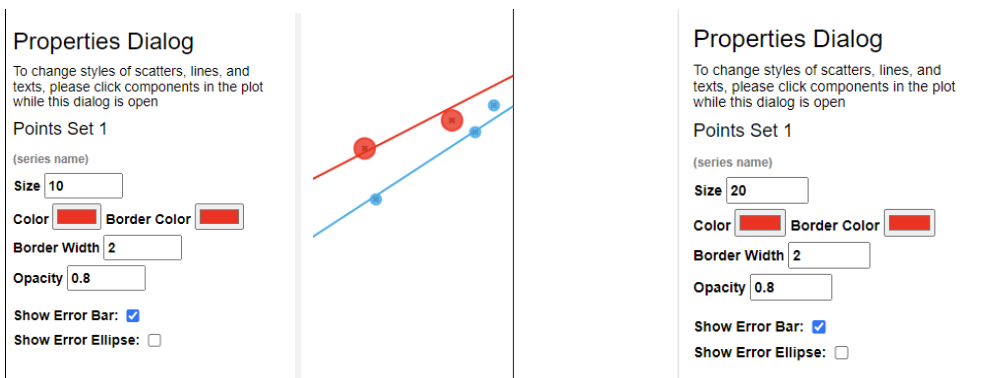


Fig. 25 Change size of set1 data points to different size

7. Save and export

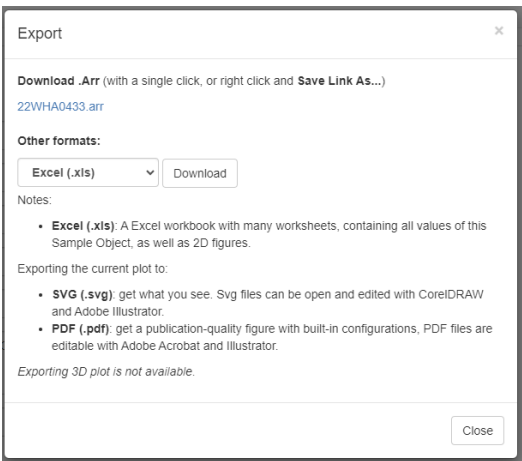


Fig. 26 Export the current object

-
- Click Export to open the Export dialog.
 - To download the arr file, click sample_name.arr or right-click to save it.
 - Excel: Contains all data and charts (except 3D plot);
 - PDF: Export 2D figures to PDF, which can be opened and edited with other software.
 - SVG: Export 2D figures to SVG.

8. Deployment

The following instruction is for local usage.

Note: This guide uses Ubuntu 22.04 as the example operating system. For Windows systems, the operations are largely similar, just skipping the Ubuntu system update steps. And one can also skip the proxy configuration steps if using the application only locally.

Update Installation Packages

```
sudo apt-get update
```

```
sudo apt-get upgrade
```

Install Dependency Libraries for gcc and g++

```
sudo apt-get install libpcre3 libpcre3-dev
```

Install PCRE Dependency Library

```
sudo apt-get install zlib1g-dev
```

Install SSL Dependency Library

```
sudo apt-get install openssl
```

Check Python 3 Version

```
python3 --version
```

```
# Expected output: Python 3.10.12
```

Install python3-pip

```
sudo apt install python3-pip
```

Create Folders, Configure Virtual Environment and Project Files

```
sudo mkdir -p /data/env
```

```
sudo mkdir -p /data/www
```

```
sudo mkdir -p /data/www/webarar # Project file directory
```

Install virtualenv

```
sudo pip3 install virtualenv
```

```
cd /data/env
```

Specify Python Version and Create a Virtual Environment Named pyweb

```
375     (/usr/bin/python3 is the Python path)
376     sudo virtualenv -p /usr/bin/python3 pyweb
377     Activate the Virtual Environment
378     source /data/env/pyweb/bin/activate # Use "deactivate" to exit the virtual environment
379     Obtain Project Files
380     sudo apt-get install git # Install Git
381     cd /data # Navigate to the target directory
382     git clone https://github.com/wuyangchn/webarar.git # Clone from repository URL (pulls to
383     current directory)
384     cp -frp webarar/* /data/www/webarar # Overwrite /data/www/webarar with /data/webarar
385     Install Python Dependencies
386     cd /data/www/webarar
387     sudo pip3 install -r requirements.txt
388     # Alternative command (skips current package if installation fails):
389     while read requirements; do sudo pip install $requirements; done < requirements.txt
390     # List of dependencies might required but not list in the requirement:
391     django
392     daphne
393     django-sslserver
394     django_redis
395     channels-redis
396     channels
397     mysqlclient
398     pandas
399     scipy
400     xlrd
401     msoffcrypto-tool
402     chardet
403     parse
404     xlswriter
405     pdf-maker
406     matplotlib
407     concurrent_log_handler
```

```
408     websockets
409     gunicorn # (optional, used for run proxy server)
410     uvicorn # (optional, used for run proxy server)
411     Configure MySQL
412     # Check MySQL version
413     mysql --version
414     # Install MySQL if not already installed
415     sudo apt-get install mysql-server # Default version: 8.0
416     # Start the service (usually running by default)
417     service mysql start
418     # Verify the service is running
419     sudo netstat -tap | grep mysql
420     # Query the current user password, copy the password
421     sudo cat /etc/mysql/debian.cnf
422     # Create dataset via Django migrate
423     cd /data/www/webbarar
424     python3 manage.py migrate
425     # Create superuser
426     python3 manage.py createsuperuser
427     Configure Redis
428     sudo apt update
429     sudo apt install redis-server
430     Configure Nginx (not required for local use)
431     # Navigate to /usr/local and download the Nginx source code:
432     cd /usr/local
433     # download nginx 1.13.7
434     sudo wget http://nginx.org/download/nginx-1.13.7.tar.gz
435     # Extract the archive:
436     sudo tar -zxvf nginx-1.13.7.tar.gz
437     # Navigate to the extracted nginx-1.13.7 folder
438     cd nginx-1.13.7/
439     # configure with module as needed
440     ./configure --prefix=/usr/local/nginx --with-http_stub_status_module --with-http_ssl_module --
```

```
441 with-http_gzip_static_moduleConfigure nginx.conf
442     make
443     make install
444     # Edit the conf file locally on Windows and upload it to Ubuntu.
445     # File path: /usr/local/nginx/conf/nginx.conf
446     # start nginx
447     cd /usr/local/nginx/sbin/
448     # First check for configuration errors:
449     ./nginx -t
450     # If no errors,
451     start Nginx: ./nginx
452     Run server
453     cd /data/www/webabar
454     # 1. Django built-in method, enough for local use
455     python manage.py runserver 8000
456     # 2. gunicorn + uvicorn method
457     python3 -m gunicorn webabar.asgi:application -k uvicorn.workers.UvicornWorker &
458
```